TEXORA · LECTURE SHEET

Material Handling Equipment

Material handling adds no value to the product — but badly handled material destroys value through damage, delay and accidents. This sheet covers the issues and importance of handling in a textile mill, how equipment is selected and classified, the conveyor family in detail, and cranes and forklifts, with a worked bucket-elevator capacity calculation.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 5 · LEVEL 4 TERM 1

1. Issues and importance

Material handling is the movement, storage, protection and control of materials through manufacture and distribution: bales to blowroom, lap and sliver between spinning stages, beams and cloth rolls in weaving, fabric batches and chemical drums in wet processing, cartons to the warehouse.

- **Cost:** handling is pure cost — it consumes labour, equipment, energy and floor space without changing the product. Reducing the number and distance of moves is direct profit.
- **Damage:** dropped bales, crushed cones, stained cloth rolls and dented beams are handling losses; the right container (can, trolley, roll stand) is a quality device.
- **Flow and WIP:** machines starve or block when transport fails; balanced handling keeps work-in-progress low and production continuous.
- **Safety:** manual lifting of bales, beams and chemical drums is a leading source of mill injuries (Topic 11); mechanisation is a safety measure first.
- **Space:** vertical handling (elevators, mezzanine conveyors) frees floor area for production.

Guiding principles: move material in the largest safe unit load; keep moves short, straight and mechanised; handle as little as possible; use gravity where possible; standardise containers; plan handling with the layout, not after it (Topic 10).

2. Selection of equipment

SELECTION CRITERION	QUESTION TO ANSWER
Material character	Bulk solid, unit load or liquid? Weight, size, fragility, temperature, corrosiveness?

Path	Horizontal, inclined or vertical? Fixed route or variable? Distance?
Rate	Required throughput (t/h, units/h) and whether flow is continuous or intermittent
Building	Floor loading, headroom, doorways, column positions
Economics	Capital cost, running and maintenance cost, labour saved, payback
Safety and environment	Guarding, dust and noise, operator exposure
Flexibility	Will routes/products change? Can the equipment adapt?

3. Classification

- **By movement:** horizontal (trucks, belt conveyors), vertical (elevators, cranes, hoists), combined (bucket elevator, overhead crane).
- **By continuity:** continuous-flow (conveyors, pneumatic systems) vs intermittent (cranes, forklifts, trolleys).
- **By drive:** manual, gravity, powered (electric, hydraulic, pneumatic).
- **By load form:** bulk-handling (conveyors, elevators, screws) vs unit-load handling (forklifts, cranes, AGVs, monorails).

4. Conveyor types

4.1 Belt conveyor

An endless belt over pulleys carries material on its top run; idlers support the belt. Simple, economical for long horizontal/inclined runs of bulk or packaged goods — bale conveyors, carton lines, coal/ash in boiler houses. Limits: incline angle, belt damage from sharp/hot loads, spillage at transfers.

4.2 Screw conveyor

A rotating helical screw inside a trough pushes material forward. Compact and enclosed (dust-tight), suited to powders and granules — dyes and chemicals dosing, fibre dust, boiler ash. Short-to-medium distances; power rises with length; can meter feed rate by speed.

4.3 Chain conveyor

Chains drag or carry the load (drag-chain, apron and slat forms). Strong, heat- and impact-resistant; used for heavy unit loads, bale lines and hot or abrasive bulk where a belt would fail.

4.4 Flight conveyor

Scraper flights attached to a chain drag bulk material along a trough (en-masse variants move it as a plug). Common for ash, dust and granular solids in enclosed troughs; simple and dust-controlled.

4.5 Bucket elevator

Buckets fixed to a belt or chain lift bulk material vertically: loaded at the boot, carried up, and discharged at the head (centrifugally or by gravity, depending on speed and design). Used for chemicals, dust/flock reclaim, grain-like solids and ash. Capacity is calculated in the worked example below.

4.6 Pneumatic conveyor

Material is suspended in an air stream inside a pipeline — dilute-phase (high velocity) or dense-phase (low velocity, high pressure). In textiles this family includes blowroom transport fans moving opened fibre between machines, flock/dust extraction, and powder-chemical transfer. Advantages: enclosed, dust-free, flexible routing, no moving parts in contact with material. Costs: high energy per tonne, wear, and the need for filtration (Topic 10).

4.7 Hydraulic conveyor

Material is carried in a water stream through pipes or flumes (slurry transport). Rare inside textile mills but used for wet waste and in process industries; included here for completeness of the classification — selection would need abundant water and separation/drying at delivery.

CONVEYOR	DIRECTION	TYPICAL TEXTILE-MILL DUTY	MAIN LIMIT
Belt	Horizontal / incline	Bales, cartons, coal	Incline angle; belt damage
Screw	Horizontal / incline	Dye & chemical powders, dust	Length; wear
Chain / apron	Horizontal / incline	Heavy bales, hot/abrasive loads	Noise; lubrication
Flight	Horizontal / incline	Ash, dust in troughs	Trough wear
Bucket elevator	Vertical	Chemicals, reclaim solids	Boot feeding; blockage
Pneumatic	Any (piped)	Opened fibre, flock, powders	Energy cost; filtration
Hydraulic	Any (piped/flume)	Wet waste slurry	Water use; separation

WORKED EXAMPLE — BUCKET-ELEVATOR CAPACITY

Bucket volume V = 4 L; bucket spacing = 0.40 m; belt speed = 1.2 m/s; bulk density of the material = 600 kg/m³. Assume buckets pass fully loaded (filling factor taken as 1 for this exercise).

Buckets passing per second = belt speed ÷ spacing = 1.2 / 0.40 = **3 buckets/s**.

Volume rate = 3 × 4 L = 12 L/s = 12 × 3.6 = **43.2 m³/h**.

> Mass rate = 43.2 m³/h × 600 kg/m³ = 25,920 kg/h ≈ **25.9 t/h**.
> In practice a filling factor (typically 0.6–0.8 for free-flowing solids, illustrative) multiplies this ideal value, and the drive is sized with a margin for starting under load.

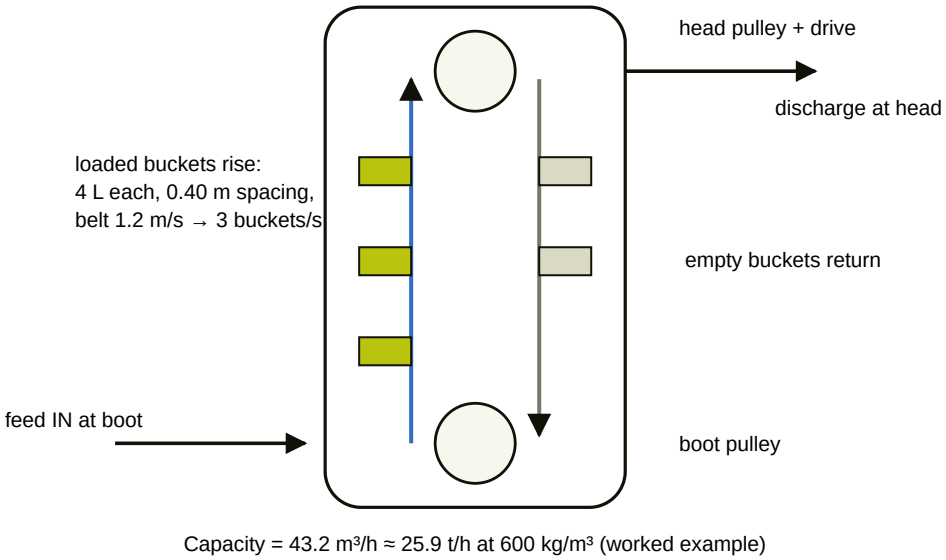


Figure 1 — Bucket elevator. Buckets on the rising leg are loaded at the boot and discharge at the head; labels repeat the worked-example values.

5. Cranes

Cranes lift and move heavy loads intermittently within a fixed area. Mill-relevant types: **overhead travelling (EOT) cranes** spanning the dye house or spinning hall — for lifting dyeing machines, calenders, heavy machine parts and beams; **jib cranes** serving one workstation; **gantry cranes** in yards and bale godowns; and **hoists/monorails** for single-path lifting. Safe working load (SWL) must be marked, lifting tackle inspected, and loads never carried over people (Topic 11).

6. Forklifts and trucks

Forklift trucks lift unit loads on forks — pallets of cartons, yarn cases, chemical drums, fabric-roll stillages — and are the most flexible handling equipment in the mill. Counterbalance trucks (electric indoors for clean air; engine-driven outdoors), hand pallet trucks for short moves, and stackers for racking cover most duties. Rules: capacity falls as load centre grows — the rating plate governs; travel with load low and tilted back; pedestrian separation in warehouses. Hand trolleys, sliver-can trains and roll trolleys remain the in-process standard between textile machines.

Glossary

Unit load — material consolidated into one handled item (pallet, can, roll). **Throughput** — quantity moved per unit time. **Bucket elevator** — vertical conveyor with buckets on belt/chain. **Pneumatic conveying** — transport of solids suspended in an air stream. **EOT crane** — electric overhead travelling crane. **SWL** — safe working load marked on lifting equipment. **WIP** — work in progress between processes.

Key points

• Handling adds cost, never value: minimise moves, mechanise heavy ones, protect the product.
• Selection follows material, path, rate, building, economics, safety and flexibility.
• Bucket-elevator example: 1.2/0.40 = 3 buckets/s × 4 L = 43.2 m^3/h ≈ 25.9 t/h at 600 kg/m^3.
• Cranes = heavy intermittent lifting in a fixed area; forklifts = flexible unit-load movement; conveyors = fixed-route continuous flow.

Practice

1. A bucket elevator has 6 L buckets at 0.5 m spacing and belt speed 1.0 m/s. Material density is 500 kg/m^3. Calculate capacity in m^3/h and t/h (filling factor 1).
2. Choose equipment, with one-line reasons, for: (a) opened cotton fibre between blowroom machines; (b) 2-tonne dyeing machine to be positioned in the dye house; (c) dye powder from store to dosing point, dust-free.
3. List four selection criteria and the failure that follows if each is ignored.

Answer key

1. Buckets/s = 1.0/0.5 = 2. Volume = 2 × 6 = 12 L/s = **43.2 m^3/h**. Mass = 43.2 × 500 = 21,600 kg/h = **21.6 t/h**.
2. (a) Pneumatic transport — enclosed ducting is the standard fibre path. (b) EOT crane — heavy intermittent lift within the hall, precise positioning. (c) Screw conveyor (enclosed) or pneumatic — dust-tight powder transfer.
3. Any four from the selection table, e.g. material character ignored → damage/blockage; rate ignored → bottleneck starving machines; building ignored → equipment cannot be installed or floor overloaded; safety ignored → lifting and dust injuries.